Assignment
Md. Mostafizur Rahman
IT 24617

Primitive Root:

A number g is called a primitive root modulo n
if g is a generator of the multiplicative group of integers modulo n
That means: $g^k \equiv 1 \pmod{n}$
only when $k = \varphi(n)$
when $\varphi(n)$ is Euler's totient function
In other words,
the smallest positive integer k for which $g^k \equiv 1 \pmod{n}$.

Example 1

Find a primitive root modulo 7

$\varphi(7) = 6$

Check possible $g = 2, 3, 4, 5$

$2^1 = 2,\ 2^2 = 4,\ 2^3 = 8 \equiv 1 \pmod{7}$ ⇒ not prim.

$3^1 = 3,\ 3^2 = 2,\ 3^3 = 6,\ 3^4 = 4,\ 3^5 = 5,\ 3^6 = 1$

Since $3^6 \equiv 1$ and no smaller power gives
3 is a primitive root mod 7